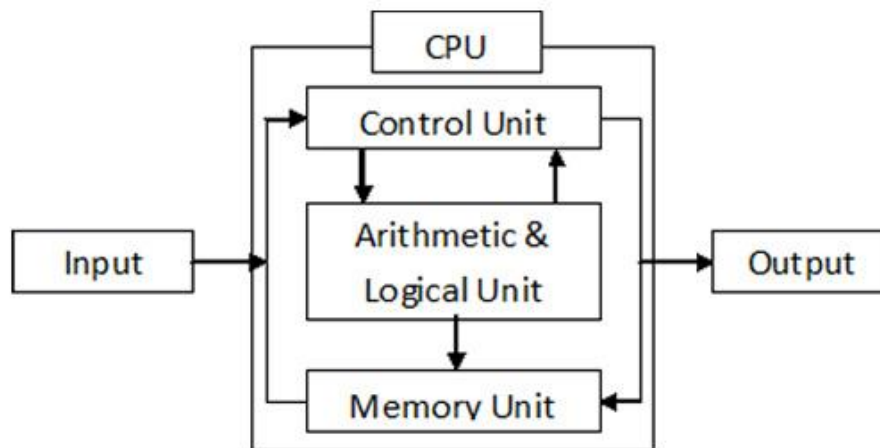


Computer Programming (Unit – I Syllabus)

History of Computers – Block diagram of a Computer – Components of a Computer system – Classification of computers - Hardware – Software – Categories of Software – Operating System – Applications of Computers – Network structure – Internet and its services – Intranet – Study of word processor – Preparation of worksheets.

University Questions – 2 Marks

1. List the various components of a computer system. (Dec 2017) (Jan 2015) (May 2014) (Dec 2019)



2. Name the different categories of software. (Dec 2017) (May 2017) (Nov 2016) (May 2016)

- System Software
 - Operating System
 - Language Translator
 - Device Driver
 - Utility Software
- Application Software
 - Database Management System
 - Presentation Application
 - Spreadsheet
 - Image Editor
 - Word Processor
 - Desktop Publishing Software, etc.,

3. Define Intranet. (Dec 2019) (Dec 2018) (May 2018) (Nov 2016)

- Intranet is a network connection within a company or organization using TCP/IP and other internet standard protocols.

- An intranet is a private network contained within an enterprise that is used to securely share company information and computing resources among employees.

4. Compare and contrast Internet and Intranet. (Sep 2020) (May 2017)

Sl.No.	Internet	Intranet
1	Internet is used to connect different network of computers simultaneously.	Intranet is used to connect computer within an organization.
2	Internet is a public network.	Intranet is a private network.
3	Anyone can access Internet.	Accessed only by the employees of that organization
4	In internet, there are multiple users.	In intranet, there are limited users.
5	Internet is unsafe.	Intranet is safe.

5. What is a word processor? Write the uses of a word processor. (May 2019) (May 2014)

Word processor is a software used to compose, format, edit and print electronic documents. It involves not only typing, but also checking the spelling and grammar of the text and arranging it correctly on a page.

Word processing software enables the user to do the following:

- To create a document
- Store it electronically on a USB or on a computer
- Display it on a screen
- Modify or format it by entering commands and characters from the keyboard
- Print the document on a printer

6. Define “Operating system”. What is the role of an operating system? (May 2018) (Jan 2015)

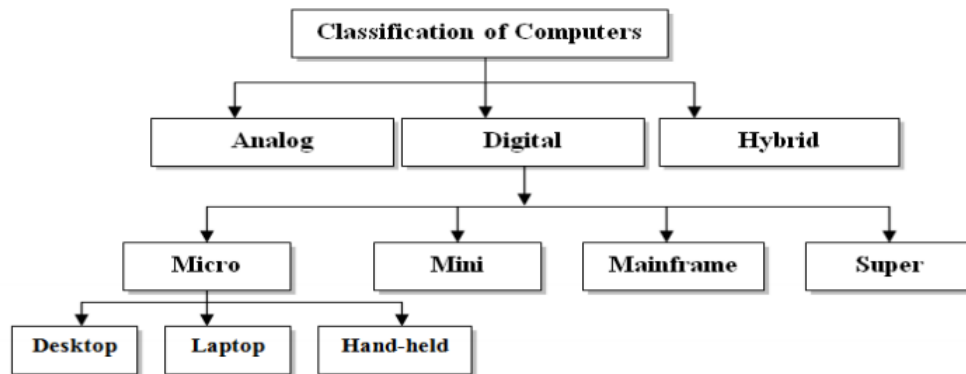
Operating system (OS) is an interface between hardware and user. Operating system is the first layer of software loaded into computer memory.

All other software that gets loaded for various common core service. In other words System Software provides the software platform; on top of which other programs can run (system software is the base for other software to run).

These common core service includes

- Disk access
- Memory management
- Configuring and controlling peripheral devices
- Managing essential file operations, including formatting or copying disks, and renaming or deleting files
- Monitoring system performance (Security management)
- Providing a user interface

7. List the different classification of computers. (Dec 2018) (Jan 2016)



8. Distinguish between analog and digital computers. (May 2016)

Analog Computer:

- Analog computer system is the very old computer system which operates on the mathematical variables in the form of continuously changeable physical quantities/entities like mechanical, electrical, hydraulic, etc.
- They use continuous values rather than discrete values so they work on analog signal.
- The advantage of analog computer is that all calculation takes place in parallel and hence it is very fast. It is used for engineering and scientific application. But their accuracy is poor as compared to digital computer.

Digital Computer

- The digital computer works with quantities represented as digits. In digital computer, both numeric and non-numeric information's are represented as strings of digits.
- In Digital Computer, the input data is discrete in nature. It is represented by binary notation in the form of 0's and 1's.
- Digital computers are much faster than analog computers and are more accurate. Digital computers are largely used for business and scientific applications

9. Differentiate Application and System software? (Jan 2016)**System Software:**

- System software consists of several programs, which are directly responsible for controlling, integrating and managing the individual hardware components of a computer system.
- System Software
 - Operating System
 - Language Translator
 - Device Driver
 - Utility Software

Application Software:

- Application software is used to accomplish specific task rather than just managing a computer system.
- It may consist of a single program, eg. Notepad.
- It may consist of a collection of programs, called software package, which work together to accomplish a task, eg. Database Management Software
- Application Software
 - Database Management System
 - Presentation Application
 - Spreadsheet
 - Image Editor
 - Word Processor
 - Desktop Publishing Software, etc.,

10. Differentiate between the first and second generation computers. (Sep 2020)**First Generation Computers:**

- These computers were based on vacuum tube technology.
- These were the fastest computing devices of their times.
- It was very large and required a lot of space for installation.
- Since thousands of vacuum tubes were used, they generated a large amount of heat.
- These machines were unreliable and prone to frequent hardware failures.

Second Generation Computers:

- These machines were based on transistor technology.
- These were smaller as compared to the first generation computers.
- The computational time of these computers was reduced to microseconds from milliseconds.
- These were more reliable and less prone to hardware failure.

11. Define a Computer. (May 2019)

- The computer is an electronic device which converts raw data into valid or meaningful information.
- A device that designated in such a way, it automatically accepts and stores input data, process them and produce the desired output.
- Computer is an automatic electronic apparatus for making calculation or controlling operations that are expressible in numerical or logical terms.

(11 Mark Questions)

1. Sketch a block diagram and explain the various components of a Computer System. (May 2019) (Dec 2018) (May 2018) (Dec 2017) (May 2017) (May 2016) (Jan 2016) (May 2014) (OR)

Illustrate the block diagram of a computer and its components.

What is a computer? Explain its characteristics and block diagram of a computer.

Explain the basic organization of computer with suitable block diagram.

Explain the fundamental units of a computer with a neat block diagram.

Draw the block schematic of a computer and explain the components.

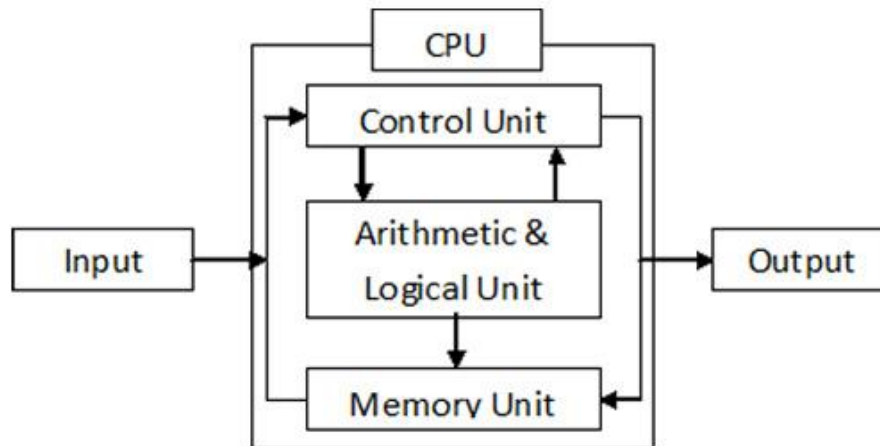
Computer:

- The computer is an electronic device which converts raw data into valid or meaningful information.
- A device that designated in such a way, it automatically accepts and stores input data, process them and produce the desired output.

Characteristics of Computer:

- **Speed:** Computers process data at an extremely fast rate – millions of instructions per second.
- **Accuracy:** Besides efficiency, computers are accurate as well. The level of accuracy depends on the instructions and the type of machine being used.
- **Diligence:** Computer, being a machine, does not suffer from the human traits of tiredness and lack of concentration.
- **Reliability:** Reliability is the measurement of performance of a computer, which is measured against some predetermined standard for operation without any failure.
- **Storage Capability:** Computers can store large amounts of data and can recall the required information almost instantaneously.
- **Versatility:** Computers are quite versatile in nature. They can perform multiple tasks simultaneously with great ease.
- **Resource Sharing:** With the tremendous growth in computer technologies, computers today have the capability to connect with each other.

Block Diagram of Computer



- A computer is an information processing machine.
- It can be viewed as a system, which consists of a number of interrelated components that work together with the aim of converting data into information.
- To attain information, data is entered through input devices.
- This data is processed using the central processing unit
- The processed data is displayed to the users using various output devices.
- Hardware of the computers
 - Input Unit
 - Output Unit
 - Central Processing Unit
 - Memory / Storage Unit
 - Arithmetic Logic Unit
 - Control Unit

Input Unit:

- Input devices are electromechanical devices that allow the user to feed information into the computer for analysis, storage and to give commands to the central processing unit.
- Data and instructions are entered into the memory of a computer through input devices.
- It captures information and translates it into a form that can be processed by the central processing unit.
- Computer accepts input in two ways, either manually or directly.
 - **Manual Data Entry:** the users enters the data into the computer by hand, for example, by using keyboard and mouse.
 - **Direct Data Entry:** information is fed into the computer automatically from a source document (like barcode).

Some commonly used input devices are **keyboard, mouse, joystick and scanner.**

Central Processing Unit:

The Central Processing Unit referred to as the brain of a computer system, converts data(input) into meaningful information(output). A CPU controls all internal and external devices, performs arithmetic and logic operations, and operates only on binary data, that is,

The CPU is sub-divided into following sub-system

- Control unit
- Arithmetic/logic unit (ALU) and
- Memory unit
 - Primary storage
 - Secondary storage

Control Unit:

- The control unit controls all other units in the computer. It directs the flow of data between Memory and Arithmetic Logic Unit. It controls and co-ordinates the entire computer system.
- It fetches program instruction from the primary storage unit, interprets them and ensures correct execution of the program.
- It also controls the input/ output devices and directs the overall functioning of the other units of the computer.

Arithmetic / Logic unit (ALU)

The task of performing operations like arithmetic and logical operations is called processing. ALU Comprises of two units. Arithmetic unit and Logic unit

Arithmetic Unit:

- Contains circuitry that is responsible for performing the actual computing and carrying out the arithmetic calculations such as addition, subtraction, multiplication and division.
- It can perform these operations at a very high speed.

Logic Unit:

- Perform logical operation based on the instruction. The operations are logical comparisons between data items.
- This unit can compare numbers, letters or special characters. Logic unit can test for 3 conditions: equal-to condition, Less-than condition and Greater-than condition.

Memory Unit

- The process of saving data and instructions permanently is known as memory.
- Data has to be fed into the system before the actual processing starts. It is because the processing speed of Central Processing Unit (CPU) is so fast that the data has to be provided to CPU with the same speed.

- Therefore the data is first stored in the memory unit for faster access and processing.
- The memory unit performs the following major functions:
 - Data and instructions required for processing (received from input devices)
 - Intermediate results of processing
 - Final results of processing, before they are released to an output device

Primary memory

- Also known as main memory.
- Used to hold running program instructions
- Used to hold data, intermediate results, and results of ongoing processing of job(s)
- Fast in operation, Expensive and Volatile
- Primary memory can be further classified into
 - RAM (Random Access Memory)
 - ROM (Read only Memory)

Secondary memory

- Also known as auxiliary memory or external memory.
- Used to hold stored program instructions
- Used to hold data and information of stored jobs
- Slower than primary storage
- Large Capacity and Non-volatile (Retains data even without power)

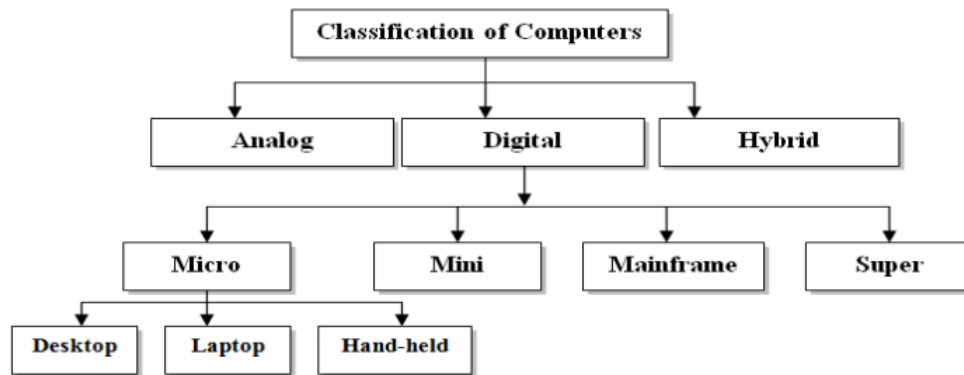
Output Unit:

- Devices used to get the response or result of a process from the computer is called output.
- This is the process of producing results from the input data.
- The outputs which can be easily understood and used by human beings are in the form of hard copy Soft copy.
 - Hard copy is the physical form of output is referred to as hard copy.
 - Soft copy is the electronic version of the output is referred as soft copy.
 - Some common output devices are Monitor, Printer, Plotter and Speaker, etc.,

2. Briefly discuss the classification of computers with their features.(Sep 2020) (Jan 2015)

Classification of Computers

- Computers are available in many sizes and types.
- Computers also differ based on their data processing abilities. Based on physical size, performance, and application areas, we can generally divide computers into four major categories: micro, mini, mainframe and super computer.



The digital computers are classified as

- Micro Computers
- Mini Computers
- Mainframe Computers
- Super Computers

Micro Computers:

- Small, low cost digital computer consists of a micro processor, a storage unit, an input channel and an output channel all of which may be on one chip inserted into one or several PC Boards.
- In addition of a power supply and connecting cables, appropriate peripherals (keyboard, monitor, printer and disk drivers).
- An operating system and other software programs can provide a complete micro computer system Examples: IBM-PC, Pentium 100, IBM-PC Pentium 200 and Apple macintosh.
- Micro computers includes
 - Desktop
 - Laptop
 - Hand-held

Desktop Computers:

- This is the computer mostly preferred by office/home users.
- These Computers lesser in cost, small ,they are Also known as Personal Computer(PC)
- We can easily arrange it to fit in our home / office with all it accommodations.
- To-day this is thought to be the most popular computer in all.
- Some of the major personal computer manufacturers are APPLE, IBM, Del.

Laptop:

- A portable computer, that is, a user can carry it around.
- Since the laptop computer resembles a notebook, it is also known as “notebook”.
- Laptop computers enclosing all the basic features of a normal desktop computers.
- Advantage:
 - Can use anywhere and at anytime, especially when one is traveling.
 - They do not need external power supply as a rechargeable battery is completely self contained in them.
- Disadvantage: Expensive compared to desktop computers.

Hand-Held Computer:

- A hand-held, also called personal digital assistant (PDA) is a computer that can conveniently be stored in a pocket (of sufficient size) and used while the user is holding it. They are slightly bigger than the common calculators.
- A PDA user generally uses a pen or electronic stylus instead of keyboard for input.
- Since these computers are easily fitted on the palm, they are also known as palmtop computers.
- They usually have no disk drive, rather they use small cards to store programs and data.
- However they can be connected to printer or a disk drive to generate output or store data.
- Examples: Apple Newton, Casio
- **Disadvantage:** It has limited memory and are less powerful as compared to desktop computers

Mini Computers:

- In the early 1960s, Digital Equipment Corporation (DEC) started shipping its PDP series computer, which the press described and referred as mini computers.
- The mini computer is a small digital computer whose process and storage capacity is lesser than that of a mainframe, but more than that of micro computer.
- Its speed of processing data is in between that of a mainframe and a micro computer.
- It is about the size of a two drawer filing cabinet.
- It is used as desktop device that is often connected to a mainframe in order to perform the auxiliary operations.
- Minicomputer (sometimes called a mid-range computer) is designed to meet the computing needs for several people simultaneously in a small to medium-sized business environment.
- It is capable of supporting from 4 to about 200 simultaneous users.
- Some of the widely used mini computers are PDP 11, IBM (8000 series) and VAX 7500

Mainframe Computer:

- A mainframe is an ultra-high performance computer made for high-volume, processor-intensive computing.
- It consists of a high end computer processor, with related peripheral devices, capable of supporting large volumes of data processing, high performance online transaction processing systems, and extensive data storage and retrieval.
- Mainframes are the second largest in capability and size of the computer family.
- Mainframe allows its user to maintain large information storage at a centralized location.
- Examples of mainframe computers are IBM's ES000, VAX 8000 and CDC 6600.

Super Computers:

- Super computers are the special purpose machines, which are specially designed to maximize the number of Floating Point Operation Per Second(FLOPS).
- Any computer below one gigaflop/sec is not considered a super computer.
- A super computer has the highest processing speed at a given time for solving scientific and engineering problems.
- It contains a number of CPUs that operate in parallel to make it faster.
- Its processing speed lies in the range of 400-10,000 millions of floating point operation per second.
- A super computer can process a great deal of information and make extensive calculations very quickly.
- Applications involve weather forecasting, military applications, electronic design etc.,
- CRAY-3, Cyber 205 and PARAM are some well known super computers.

3. Illustrate in detail about internet and its services.(Jan 2016) (May 2019) (Dec 2018) (May 2017) (Nov 2016) (Jan 2015) (May 2014) (OR)

List the various internet services available in the world and explain any three of the services in detail.

Discuss the various services for which you use Internet.

Identify the various Internet services available in the globe and explain any four services in detail.

Describe how computers connected to the Internet communicate and explain briefly two internet services.

World Wide Web:

- A system of Internet servers that support specially formatted documents.

- The documents are formatted in a markup language called HTML (HyperText Markup Language) that supports links to other documents, as well as graphics, audio, and video files.
- This means you can jump from one document to another simply by clicking on hot spots.
- Not all Internet servers are part of the World Wide Web. There are several applications called Web browsers that make it easy to access the World Wide Web;
- Two of the most popular being Firefox and Microsoft's Internet Explorer.

e-mail:

- Electronic mail is the transmission of messages over communications networks.
- The messages can be notes entered from the keyboard or electronic files stored on disk.
- Most mainframes, minicomputers, and computer networks have an e-mail system. Some electronic-mail systems are confined to a single computer system or network, but others have gateways to other computer systems, enabling users to send electronic mail anywhere in the world.
- Companies that are fully computerized make extensive use of e-mail because it is fast, flexible, and reliable.
- Sent messages are stored in electronic mailboxes until the recipient fetches them.
- All online services and Internet Service Providers (ISPs) offer e-mail, and most also support gateways so that you can exchange mail with users of other systems.

File transfer Protocol (FTP)

- File Transfer Protocol, the protocol for exchanging files over the Internet.
- FTP works in the same way as HTTP for transferring Web pages from a server to a user's browser and SMTP for transferring electronic mail across the Internet in that, like these technologies, FTP uses the Internet's TCP/IP protocols to enable data transfer.
- FTP is most commonly used to download a file from a server using the Internet or to upload a file to a server (e.g., uploading a Web page file to a server).

Telnet

- A terminal emulation program for TCP/IP networks such as the Internet. The Telnet program runs on your computer and connects your PC to a server on the network.
- You can then enter commands through the Telnet program and they will be executed as if you were entering them directly on the server console.
- This enables you to control the server and communicate with other servers on the network.
- To start a Telnet session, you must log in to a server by entering a valid username and password. Telnet is a common way to remotely control Web servers.

Internet Relay Chat (IRC):

- Internet Relay Chat, a chat system developed by Jarkko Oikarinen in Finland in the late 1980s.
- IRC has become very popular as more people get connected to the Internet because it enables people connected anywhere on the Internet to join in live discussions.
- Unlike older chat systems, IRC is not limited to just two participants.
- To join an IRC discussion, you need an IRC client and Internet access.
- The IRC client is a program that runs on your computer and sends and receives messages to and from an IRC server.
- The IRC server, in turn, is responsible for making sure that all messages are broadcast to everyone participating in a discussion. There can be many discussions going on at once; each one is assigned a unique channel.

Usenet:

- USENET is a worldwide bulletin board system that can be accessed through the Internet or through many online services.
- The USENET contains more than 14,000 forums, called newsgroups, which cover every imaginable interest group. It is used daily by millions of people around the world.

Chatting and Instant Messaging:

- Chat programs allow users on the internet to communicate with each other by typing in real-time.
- Note that even though chatting is based on standardized IRC system, unlike the user does not need to have any special software to connect to any chat rooms.
- A variation of chat is instant messaging where a user on the web can contact another currently logged in and type a conversation.
- To avail this Internet service, the user must have software called instant messenger installed on the system.

Internet Telephony:

- Internet telephony is the use of Internet rather than traditional telephone company infrastructure, to exchange spoken or other telephonic information.
- It consists of hardware and software that enable people to use the Internet as a transmission medium phone call.
- There are many Internet telephony applications available. Some, such Cool Talk and NetMeeting come bundled with popular web browsers.

Video Conferencing:

- Video conferencing uses the same technology as IRC, but also provides sound and video pictures. It enables direct face-to-face communication across network.

Commerce through Internet:

- Today business is taking place through electronic telecommunication media. Nowadays, various organizations offer the facility of shopping online. This type of business model is known as Electronic Commerce or E-commerce.
- E-commerce refers to buying and selling goods and online. When discussing about e-commerce, we normally talk about virtual shopping malls, up stores where the user virtually visits and selects the product(s) of his/her choice. After selecting the desired product(s), the payment for the purchase is done.
- Online payment can be through credit card. Some of the well-known e-commerce sites are www.amazon.com and www.ebay.com. Mobile commerce or M-commerce refers to transactions through mobile phone network and data connection that result in the transfer of value in exchange for goods and services.

4. Describe about various applications of Computers. (May 2018) (Dec 2017) (May 2014)

APPLICATIONS OF COMPUTERS / ROLE OF COMPUTERS

Computer is used everywhere in the world in every field of life. There are many applications of computer following list demonstrate the various Applications of computer in today's arena.

Science:

- Scientists have been using computers to develop theories, to analyse, and test the data.
- The fast speed and accuracy of the computer allow different scientific analyses to be carried out.
- They can be used to generate detailed studies of how earthquakes affect buildings or pollution affects weather pattern.

Education:

- Computers have also revolutionized the whole process of education.
- Currently, the classrooms, libraries and museums are utilizing computers to make education much more interesting.
- Computer Aided Education and Computer Based Training packages are making learning much more interactive.
- Educational process leads to improve student's performance in thinking logically, formulating problem solving procedures and understanding relationship.

Medicine and Health Care:

- There has been an increasing use of computers in the field of medicine.
- Doctors are using computers right from diagnosing the illness to monitoring a patient's status during a complex surgery.
- By using automated imaging techniques, doctors are able to look inside a person's body and can study each organ in detail (CAT scans or MRI scans)

Engineering/Architecture/Manufacturing:

- The architects and engineers are extensively using computers in designing and drawings.
- Computers can create objects that can be viewed from all the three dimensions. By using techniques like virtual reality, architects can explore houses that have been designed but not built.
- The Computer Aided Manufacturing is used in designing products, ordering parts, and planning production.

Entertainment:

- Computers are finding greater use in entertainment industry.
- They are used to control the images and sounds. Special effects, which mesmerize the audience, would not have been possible without computers.
- The computerized animation and colorful graphics have modernized the film industry.

Communication:

- E-mail is one of the communication media in which computer is used.
- Through e-mail, messages and reports are passed from one person to one or more persons.
- The benefit of e-mail service is that while transferring the messages it saves time, saves paper.
- The person who is receiving the messages can read the messages whenever he is free and can save it, reply it, forward it or delete it from the computer.

Publishing:

- Computers have created a field known as DTP (Desktop Publishing).

Governance:

- IT governance derives from corporate governance and deals primarily with the connection between Business focus and IT management of an organization.

Business:

- The computer characteristics as speed Accuracy, Automatic, Endurance, versatility, Reduction of cost has made it an integrated part in all Business Organization.
 - Computer used in Business Organization for
 - Payroll calculation
 - Budgeting
 - Managing employee database
 - Maintenance of stocks etc...

Military:

- Computers are largely used in Defense. Some of the Military areas where a computer has been used are:
 - Radar Monitoring
 - Military Communication
 - Missile Control
 - Smart Weapons

Banking:

- Banks provide following facilities:
 - Bank on-line accounting facility, which includes Current Balance, Deposits, Shares, Trustee Records.
 - ATM machine are making it even easier for customers to deals with banks

Marketing:

- In Marketing uses of computer are following
 - Advertisement
 - At Home Shopping

5. Write briefly about the software and hardware of a computer.(Nov 2016) (May 2019) (May 2018)**SOFTWARE :**

Software refers to the instructions or programs that tell the hardware what to do. It is responsible for controlling, integrating and managing the hardware components of a computer and to accomplish specific tasks.

Software can be categorized in to

- System Software
- Application Software

SYSTEM SOFTWARE:

- System software acts as an interface between the hardware of the computer and the software applications. The purpose of this software is to help the user to run the computer system.
- System software consists of several programs, which are directly responsible for controlling , integrating and managing the individual hardware components of the computer system.
- Some system software are
 - Operating System
 - Device Drivers
 - Language Translators
 - System Utilities

Operating System:

- Operating system (OS) is an interface between hardware and user. Operating system is the first layer of software loaded into computer memory.
- All other software that gets loaded for various common core service. In other words System Software provides the software platform; on top of which other programs can run (system software is the base for other software to run).
- Operations of Operating System
 - Process Management
 - Memory Management
 - File management
 - Device Management
 - Security management
 - User Interface

Device Driver Software:

- Device driver or software driver are system programs which are responsible for proper functioning of devices.
- Every device such as printer, monitor, mouse or keyboard, has a driver associated with it for its proper functioning.
- Whenever a new device is added to the computer system, a new device driver must be installed before the device is used.
- In modern operating system, most hardware drivers such as keyboard driver come with the operating system

- Eg. When a user want to print a document, the processor issues a set of generic commands to the printer driver, and the driver translate those commands into specialized instructions that the printer can understand.

Language Translator

- Computer can understand a language consisting of 0s and 1s called machine language
- Special programming languages called high- level language were developed that resemble natural languages like English.
- Therefore, tool was required which could translate a program written in a programming language to machine language.
- Depending on the programming language used, language translators are divided into three major categories.
 - Compiler
 - Interpreter
 - Assembler

Compiler:

- The program written in any high level language is converted into machine language using a compiler. Compiler translate source code (user written program) into object code (binary form)

Interpreter:

- Interpreter analyses and executes the source code in line-by-line manner, without looking at the entire program. The advantage of interpreter is that they can execute a program spontaneously.
- Compiler requires some time before an executable program is formed because it looks at the whole source code.

Assembler:

- Assembly language is closest to the machine code. Assembly language is fundamentally a symbolic representation of machine code.
- The assembly language program must be translated into machine code by a separate program called an assembler.

System Utility:

- System utility program perform day to day task related to the maintenance of the computer system.
- They are used to support, enhance and secure existing programs and data in the computer system.

- Some utility programs are usually provided along with the operating system, some are free while some need to be purchased from the 3rd party vendors.
- Eg. For system utilities:
 - Disk checkers
 - Disk cleaners
 - Antivirus utilities
 - Registry cleaners
 - Network managers

APPLICATION SOFTWARE:

- Application software is used to accomplish specific tasks. It consist of a single program, such as Microsoft's Notepad(for writing and editing simple text).
- It may also consist of a collection of programs, often called a software package, which work together to accomplish a task, such as database management software.
- Application software is dependent on system software. Application software is controlled by system software, which manages hardware devices and performs back ground tasks for them.
- Application software ranges from games, calculators and word processors(text editor), paint(image Editor).
- Some of the most commonly used application software are
 - Word processors
 - Spreadsheets
 - Image editors
 - Database management systems
 - Presentation Applications
 - Desktop Publishing Software

Word Processors:

- Word processor is a software used to compose (create), format, edit and print electronic document.
- It involves not only typing, but also checking the spelling and grammar of the text and arranging it correctly on a page.
- Example of some well known word processors are Microsoft Word and WordPerfect.

Spreadsheets:

- One of the first commercial uses of computers was in processing payroll and other financial records.
- A Spreadsheet application is a rectangular grid, which allows text, numbers and complex functions to be entered into a matrix of thousands of individual cells.

- Examples are Microsoft Excel and Lotus 1-2-3

Image Editors:

- Image editors are designed specifically for capturing, creating, editing and manipulating images.
- Most image editors provide a variety of special features for creating and altering images. Examples are Adobe photoshop , CorelDraw, etc.,

Database Management System (DBMS):

- Database management software is a collection of computer programs that allow storage, modification and extraction of information form a database in an efficient manner.
- It provides tools for data input, verification, storage, retrieval, query and manipulation. It also controls the security and integrity of the database from unauthorized access. Examples are Ms Access, FoxPro, Oracle etc.,

Presentation Applications

- Presentation graphics software allows users to create computerized slide shows that combine text, numbers, animation, graphics, sounds, and videos.
- A slide is an individual document that is created in presentation graphics software. A slide show may consist of any number of individual slides Example Ms Powerpoint Presentation.

Desktop Publishing (DTP) software:

- Used to describe the creation of printed documents using a desktop computer.
- It's a technique of using a personal computer to design images and pages and assemble type and graphics then using a laser printer or image setter to output the assembled pages onto paper, film or printing plate.
- This software is used for creating magazines, books, newsletters and so on. Example: Quark Express and Adobe PageMaker, Adobe InDesign.

6. Write in detail about the history of Computers. (Dec 2019)**HISTORY OF COMPUTERS (EVOLUTION OF COMPUTERS)****Abacus**

- In the beginning when the task was simply **counting or adding**, people used either their fingers or pebbles along lines in the sand. In order to simplify the process of counting, people in Asia built **a counting device called abacus**.
- This device allowed users to do calculations using a system of **sliding beads arranged on the rack**.
- The abacus was simple to operate and was use worldwide for centuries.

Napier Device

- **Napier invented** logarithms, **which are a technology that allows** multiplication to be performed via addition.
- **The magic ingredient is the logarithm of each operand, which was originally obtained from a printed table.**
- **But Napier also invented an alternative to tables, where the** logarithm values were carved on ivory sticks **which are now called** Napier's bones.

Slide Rule

- Napier's invention led directly to the slide rule, first built in England in 1632 and still in use in the 1960's by the NASA engineers of the Mercury, Gemini, and Apollo programs which landed men on the moon.
- It can perform all **arithmetic and trigonometric** functions.

Pascal Calculating Machine (Pascaline)

- In 1664, a French mathematician, Pascal invented the **first functional automatic calculator.**
- The brass rectangular box is called a **Pascaline**, which used eight movable dials to add sums eight figures long.

Stepped Reckoner - Leibnitz's improved Pascal Machine

- In 1694, German mathematician Gottfried wilhem von leibniz extended pascal design to perform **multiplication, division and to find square root.** This machine is known as **Stepped Reckoner.**
- It was the first mass-produced calculating device, which was designed to perform multiplication by repeated addition.

Punched Card Machine

- In 1801, a French weaver **Joseph Marie Jacquard** invented the first punched card machine.
- Jacquard was involved in designing clothes and making patterns or designs on cloth. In order to control his textile looms, he invented punched cards.
- These cards with holes were used to make attractive designs on the fabric. The presence and absence of the punched hole in the card represented two states for raising or lowering of wrap thread.
- Only these threads whose guiding hook encountered a hole in the card could enter the pattern.

Charles Babbage's Engines

- In 1822 **Charles Babbage** proposed a machine to perform differential equations called a **difference Engine.**
- This machine had a stored program and could perform calculations and print the results automatically.

- In 1833, he concentrated on **analytical engine**. The basic design of this engine included **input devices** in the form of perforated cards containing operating instructions and a store for memory of 1000 numbers up to 50 decimal digits.
- It also contains a **control unit** that allowed processing instruction in any sequence, and **output device** to produce results.
- Although the analytical engine was never constructed, it outlined the basic elements of a modern computer.

The First Electronic Computer (ABC)

- **Dr.Jhon Vincent Atanasoff** and **Clifford Berry** developed the first electronic computer and called it as **ABC – Atanasoff – Berry Computer**.
- This used **Vaccum tubes** for storage, arithmetic and logical functions. This was a special purpose machine used to solve simultaneous equations.

Electronic Numerical Integrator and Calculator (ENIAC)

- This computer used **electronic vacuum tubes** to make the internal parts of the computer.
- It embodied almost all the components and concepts of today's high speed, electronic digital computers.

Electronic Discrete Variable Automatic Computer (EDVAC)

- It was the first electronic computer to use the **stored program concept** introduced by John Von Neumann in 1946.
- It also had the capability of **conditional transfer of control**, that is, the computer could stop anytime and then resume operations.

Electronic Delay Storage Automatic Calculator

- **Electronic Delay Storage Automatic Calculator**. In 1949 at the Cambridge University EDSAC was developed. This is based on John Von Neumann's stored program concept.
- This machine used **mercury delay lines** for memory and **vacuum tubes** for logic.

Universal Automatic Computer (UNIVAC-I)

- UNIVAC – **UNIVersal Automatic Computers**. The Eckert- Mauchly Corporation manufactured UNIVAC in 1951. This implementation marked the real beginning of the computer era.
- UNIVAC was also the first computer to **employ magnetic tape**.

Microprocessors

- In 1970's the trend shifted towards a larger range of applications for cheaper computer systems. During this period many business organizations adopted computers for their offices.
- A microprocessor (uP) is a computer that is fabricated on an **integrated circuit (IC)**.
- Microprocessor, a **processors on a single Integrated Circuit (IC) Chip**.

7. Enumerate the various features of word processor with illustrations.(Sep 2020)**Word Processing:**

Word Processing is using a computer to create, edit, and print documents. Of all computer applications, word processing is the most commonly used. To perform word processing - a computer, word processing software and a printer are required.

Word processing software enables the user to do the following:

- To create a document
- Store it electronically on a USB or on a computer
- Display it on a screen
- Modify or format it by entering commands and characters from the keyboard
- Print the document on a printer

Features of Word Processor:

Word processing typically implies the presence of text manipulation functions that extend beyond a basic ability to enter and change text.

Text Manipulation:

- Describe how to change the appearance of a document by text manipulation.
- Text manipulation usually refers to the ability to change words, sentences and paragraphs which have been typed.
- The manipulation can involve the changing of the characters (letters and symbols), adding line breaks, direction of the text and changing the case (CAPITAL letter or small letter) of the characters.

Document Formatting:

- Describe document formatting. The text within a document can be formatted in terms of how the actual text appears.
- Formatting improves the readability of documents. A useful function with Word is that of the Styles which allow text to be automatically formatted allowing consistent formatting of text within body, titles, and subtitles of a document.

Graphics:

- Describe the uses of graphics in a word document Pictures, images or graphics are a very useful way of brightening up a document and making it more appealing and interesting for the end user.
- Graphics can be added to Word in a variety of ways, including inserting; Pictures, Clip Art, Shapes and SmartArt.

- When images are added to a document it is important to consider the graphic formatting options available, such as cropping, wrapping, styles, effects and position

Tabs:

- Understand the uses of tabs e.g. decimal, left, right and centre tabs.
- Tabs allow the setup of text on a page in the exact position it should appear in, by allowing the cursor to advance to the next stop or set position.
- This is achieved by setting up tabs and using the tab key (to the left of the Q key) on the keyboard, to jump from one position to the next, instead of having to use the space bar again and again and again!

Tables:

- Describe the uses of tables. A table is a grid of cells arranged in rows and columns.
- It can be useful to present data in a table inside a word processing document because it is displayed in an organised and easy to read format.
- Tables can be sorted, formatted, merged, aligned, shaded and coloured.

Mail Merge and Label Printing:

- Describe the uses of mail merge and the benefits of label printing Mail merge is the batch mailing of letters - using a letter template and a data file or address database.
- Instead of a business or college creating multiple copies of the same letter to send out in the post, one letter and one data file is created.
- The letter and data file are merged together - which means the names and address are automatically added to each of the letters and envelopes from the database or data file.

Other Features:

- Spell checking (actually checks against wordlists),
- Grammar Checking (checks for what seem to be simple grammar errors)
- Thesaurus function (finds words with similar or opposite meanings Shift F7).
- Collaborative editing, which allows two people to work on one document, automatically leaving each other notes to show where editing has taken place.
- Templates are pre-created documents which can be used to easily create documents.

Short Cut Keys:

- Ctrl + A -- Select all contents of the page.
- Ctrl + B -- Bold highlighted selection.
- Ctrl + C -- Copy selected text.
- Ctrl + X -- Cut selected text.

- Ctrl + N -- Open new/blank document.
- Ctrl + O -- Open options.
- Ctrl + P -- Open the print window.
- Ctrl + F -- Open find box.

8. What is worksheet? Describe any ten functions available in worksheet. (Dec 2019) (May 2016)

- The term **Worksheet** used in Excel documents is a collection of cells organized in rows and columns.
- It is the working surface you interact with to enter data. a workbook contains several worksheets with related content and only one of the worksheets is active at a time.

The main characteristics of worksheets:

Cells: Each worksheet has a significant number of cells that can be assigned values and various formatting options. The Cells property allows you to insert, manipulate and delete worksheet cells.

Rows and Columns: Cells in a worksheet are organized in rows and columns. The API of a worksheet allows you insert, manipulate and delete rows and columns. Additionally, you can adjust the height of specific rows and the width of columns.

Find and Replace: You can easily find and replace text and numbers in the content of a worksheet.

Protection: You can restrict the user from modifying the content and structure of the worksheet. When a worksheet is protected the user can edit the contents only of the cells that were explicitly marked as unlocked.

Some important functions in worksheets:

- **SUM:** This function adds all the values of the cells in the argument.
=SUM(number1, [number2], ...)
- **AVERAGE:** This function determines the **average** of the values included in the argument. It calculates the sum of the cells and then divides that value by the number of cells in the argument.
=AVERAGE(number1, [number2], ...)
- **COUNT:** This function counts the number of cells with numerical data in the argument. This function is useful for quickly counting items in a cell range.
=COUNT(value1, [value2], ...)
- **MAX:** This function determines the highest cell value included in the argument.
=MAX(number1, [number2], ...)

- **MIN:** This function determines the lowest cell value included in the argument.
=MIN(*number1*, [*number2*], ...)
- **IF:** The IF function checks whether a condition is met, and returns one value if true and another value if false.
=IF(*logical_test*, [*value_if_true*], [*value_if_false*])
- **ROUND:** This function rounds the given number to the specified decimal values.
=ROUND(*Cell No*, *Number of Decimal values*)
- **UPPER:** It converts the string in the given cell into upper case alphabets.
=UPPER(*Cell No*)
- **LOWER:** It converts the string in the given cell into lower case alphabets.
=LOWER(*Cell No*)
- **COUNTIF:** COUNTIF counts all cells that meet certain criteria
=COUNTIF(*Cell Range*, *Condition*)

9. Discuss the following internet terminologies. (May 2016)

- a. Band width**
- b. FTP**
- c. IP Address**
- d. Modem**
- e. Web browser.**

Bandwidth:

- Bandwidth is measured as the amount of data that can be transferred from one point to another within a network in a specific amount of time. Typically, bandwidth is expressed as a bitrate and measured in bits per second (bps).
- The term bandwidth refers to the transmission capacity of a connection and is an important factor when determining the quality and speed of a network or the internet connection.

File Transfer Protocol (FTP):

- FTP stands for File transfer protocol.
- FTP is a standard internet protocol provided by TCP/IP used for transmitting the files from one host to another.
- It is mainly used for transferring the web page files from their creator to the computer that acts as a server for other computers on the internet.

- It is also used for downloading the files to computer from other servers.

Objectives of FTP

- It provides the sharing of files.
- It is used to encourage the use of remote computers.
- It transfers the data more reliably and efficiently.

Internet Protocol:

- IP (Internet Protocol) Address is an address of your network hardware. It helps in connecting your computer to other devices on your network and all over the world. An IP Address is made up of numbers or characters.
- All devices that are connected to an internet connection have a unique IP address which means there's a need of billions of IP addresses. This requirement is fulfilled by the new IP version IPv6.

Modem:

- Modem is short for "Modulator-Demodulator." It is a hardware component that allows a computer or another device, such as a router or switch, to connect to the Internet.
- It converts or "modulates" an analog signal from a telephone or cable wire to digital data (1s and 0s) that a computer can recognize.
- Similarly, it converts digital data from a computer or other device into an analog signal that can be sent over standard telephone lines.

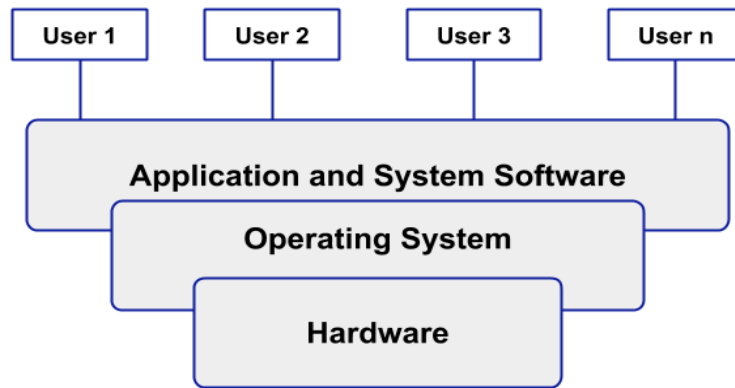
Web Browser:

- A web browser, or simply "browser," is an application used to access and view websites. Common web browsers include Microsoft Internet Explorer, Google Chrome, Mozilla Firefox, and Apple Safari.
- The primary function of a web browser is to render HTML, the code used to design or "mark up" webpages.
- Each time a browser loads a web page, it processes the HTML, which may include text, links, and references to images and other items, such as cascading stylesheets and JavaScript functions.
- The browser processes these items, then renders them in the browser window

10. Explain the functions of operating system.(May 2014)**Operating System:**

- An operating system is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.

- An operating system is a software which performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers.
- Some popular Operating Systems include Linux Operating System, Windows Operating System, VMS, OS/400, AIX, z/OS, etc.



Following are some of **important functions of an operating System**.

- Memory Management
- Processor Management
- Device Management
- File Management
- Security
- Control over system performance
- Job accounting
- Error detecting aids
- Coordination between other software and users

Memory Management

Memory management refers to management of Primary Memory or Main Memory. Main memory is a large array of words or bytes where each word or byte has its own address.

Main memory provides a fast storage that can be accessed directly by the CPU. For a program to be executed, it must be in the main memory. An Operating System does the following activities for memory management –

- Keeps tracks of primary memory, i.e., what part of it are in use by whom, what part are not in use.
- In multiprogramming, the OS decides which process will get memory when and how much.
- Allocates the memory when a process requests it to do so.

- De-allocates the memory when a process no longer needs it or has been terminated.

Processor Management

In multiprogramming environment, the OS decides which process gets the processor when and for how much time. This function is called **process scheduling**. An Operating System does the following activities for processor management –

- Keeps tracks of processor and status of process. The program responsible for this task is known as **traffic controller**.
- Allocates the processor (CPU) to a process.
- De-allocates processor when a process is no longer required.

Device Management

An Operating System manages device communication via their respective drivers. It does the following activities for device management –

- Keeps tracks of all devices. Program responsible for this task is known as the **I/O controller**.
- Decides which process gets the device when and for how much time.
- Allocates the device in the efficient way.
- De-allocates devices.

File Management

A file system is normally organized into directories for easy navigation and usage. These directories may contain files and other directions.

An Operating System does the following activities for file management –

- Keeps track of information, location, uses, status etc. The collective facilities are often known as **file system**.
- Decides who gets the resources.
- Allocates the resources.
- De-allocates the resources.

Following are some of the important activities that an Operating System performs –

- **Security** – By means of password and similar other techniques, it prevents unauthorized access to programs and data.
- **Control over system performance** – Recording delays between request for a service and response from the system.

- **Job accounting** – Keeping track of time and resources used by various jobs and users.
- **Error detecting aids** – Production of dumps, traces, error messages, and other debugging and error detecting aids.
- **Coordination between other softwares and users** – Coordination and assignment of compilers, interpreters, assemblers and other software to the various users of the computer systems.